XRPL Validator — Daily Engineering Report · 2026-11-05

Period: 2026-11-05 00:00 to 24:00 US Eastern (2026-11-05 05:00 to 2026-11-06 05:00 UTC); ledgers #108,124,749 to #108,146,083. Every count from the ledger and the validator's logs stops at #108,146,083, closed 04:59:51 UTC. Generated 2026-11-06 00:20 EST by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Status

Normal — coverage outside planned maintenance 99.991% (objective 99.9%), every validation agreed with the network, no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 21,333 of the 21,335 ledgers the network closed (99.991%); every validation matched the network's validated ledger. Unsigned: 0 during planned restarts, 2 outside maintenance (1 incident).
- **Native engine:** 1 engine receipt: #108,137,503 — fixed: commit f81491e (2026-11-05).
- **Changes:** 0 deployments, 0 commits merged into main.
- **Network:** 8 slow-close periods; the longest, 23:10 UTC–23:13 UTC, closed ledgers up to 20 s apart.
- **Independence program:** 8 of 30 qualifying days (snapshot at 05:20:43 UTC).

Service levels

Measure	Value
Ledgers closed by the network	21,335
Validations issued	21,333
Coverage (validations / ledgers closed)	99.991%
Validations matching the network's validated ledger hash	21,333 of 21,333
Unsigned during planned restarts	0
Unsigned outside maintenance	2
Coverage outside planned maintenance (objective 99.9%)	99.991%
Restarts	0
Basis: our own ledger hash	21,318
Basis: our own hash, witnessed by the reference node (quorum short)	15
External agreement, data.xrpl.org (UTC day 2026-11-05)	0.99735 (52 missed of 19,652)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#108,124,749–#108,146,083), by step:

Step	Ledgers with a receipt
Step 1 · native engine	1
Step 2 · records	1
Step 3 · order	1
Step 4 · peer data	0
Step 4 · validations	17
Step 5 · own hash	1
Writer	1

Live counters, snapshot at 05:20:43 UTC (since the last restart, 2026-11-05 00:49 UTC; not cut off at the period's last ledger):

Step	Checked	Result
1 · native engine vs libxrpl	25,182 ledgers, 6,075,853 transactions	25,181 fully matched; results 0, bytes 3, keys 0/0
2 · transaction records	25,391 ledgers, 6,146,463 records	6,146,426 exact; differ 37, tree 1
3 · transaction order	25,434 ledgers (mode: on)	25,433 exact; order 1, results 0, bytes 0, refused 1
4 · ledger data from peers	25,441 ledgers fetched	0 matched; differ 0, failed 0
4 · trusted validations	25,435 ledgers judged	25,418 agreed; differ 0, short of quorum 17
5 · our own ledger hash	25,438 ledgers	25,389 own; differ 1, partial 48
Writer	every ledger written	mismatch 0, distrusted 0

"Partial" ledgers are those the engine could not yet build alone while warming up after a restart.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	19:20 UTC	#108,137,501–#108,137,506 (6)	Step 4 · validations	16–18 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (4 of 6 signed); see INC-01
R-02	19:20 UTC	#108,137,503	Step 1 · native engine, Step 2 · records, Step 3 · order, Step 5 · own hash, Writer	Payment tx 48B0042A0F05: RippleState 12CB8DFB (modified on the network): FinalFields.Balance ours= {"currency":"524C55534400000000000…	fixed: commit f81491e (2026-11-05); awaiting merge and deployment; see INC-01
R-03	23:11 UTC	#108,140,914–#108,140,920 (7)	Step 4 · validations	16 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network

ID	Time	Ledger(s)	Step	Description	Disposition
					validated these ledgers normally (all signed)
R-04	03:59 UTC	#108,145,192–#108,145,195 (4)	Step 4 · validations	19–21 of the 28 trusted validations needed reached the validator in time	explained: network delivery, not an engine defect; the network validated these ledgers normally (all signed)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-11-05-01, INC-01 is INC-2026-11-05-01.

Incidents

INC-01 · 2 ledgers unsigned

Window	19:20:50 UTC to 19:21:00 UTC
Ledgers	#108,137,503–#108,137,504 (2)
Impact	the validator issued no validation for these ledgers; no incorrect validation was issued
Related receipts	R-01, R-02
Cause	Network-wide slow closes (19:20 UTC–19:20 UTC: up to 20 s apart) during a burst of Batch transactions (#108,137,502: 1,304 of 2,358 transactions). Step 3 salts our canonical order with the consensus set more than half the trusted validators named (sfConsensusHash), else with the root of the ledger's own transactions. On mainnet #108137501-505 (5 Nov) the trusted validators split between two ledgers: on #108137503, 17 of 35 validated it, 15 another, 1 a third. No set had more than half, the own root was not the set (it held a transaction the ledger never filed), so the order differed at position 0 and the ledger and the next went unsigned.
Action	commit f81491e — fix(order): in a validator split, seed the order with the set this ledger's own validators named; awaiting deployment

Change log

Deployments

No deployments.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Independence program

Qualifying days: **8 of 30** (snapshot at 05:20:43 UTC). A UTC day qualifies when every validation is of a ledger our own engine built, with our own transaction order and ledger data taken from peers, and coverage is at least 98%.

UTC day 2026-11-05 — tracked #108,120,209–#108,141,634	Value
Validations	21,200 (21,200 own, 0 borrowed)
Coverage of the tracked ledgers	98.95%
Our own transaction order	on
Ledger data from peers	yes

Meets every criterion.

The validator saves this record every 64 validations, so a day in progress lags the signing log slightly; it is not cut off at the period's last ledger.

Network

Measure	Value
Ledgers closed	21,335 (#108,124,749–#108,146,083)
Mean close interval	4.05 s
Longest close interval	49 s (before #108,145,193)
Closes at least 15 s apart	15
Empty ledgers	0

- Slow closes 15:13:42 UTC–15:14:00 UTC: ledgers #108,133,977, 1 close at least 15 s apart, up to 18 s.
- Slow closes 15:14:04 UTC–15:14:20 UTC: ledgers #108,133,982, 1 close at least 15 s apart, up to 16 s.
- Slow closes 16:01:21 UTC–16:01:40 UTC: ledgers #108,134,650, 1 close at least 15 s apart, up to 19 s.
- Slow closes 19:20:20 UTC–19:20:40 UTC: ledgers #108,137,502, 1 close at least 15 s apart, up to 20 s.
- Slow closes 19:21:20 UTC–19:21:40 UTC: ledgers #108,137,507, 1 close at least 15 s apart, up to 20 s.
- Slow closes 23:10:40 UTC–23:13:00 UTC: ledgers #108,140,914–#108,140,923 (10), 6 closes at least 15 s apart, up to 20 s.
- Slow closes 23:13:03 UTC–23:13:20 UTC: ledgers #108,140,927, 1 close at least 15 s apart, up to 17 s.
- Slow closes 03:59:51 UTC–04:01:20 UTC: ledgers #108,145,193–#108,145,198 (6), 3 closes at least 15 s apart, up to 49 s.

From ledger close times, which the network records at a 10-second resolution or coarser, so single intervals are approximate.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

No amendment held a majority.

Amendments enabled on mainnet: 98.

Open items

- Receipt R-02 (#108,137,503): fixed: commit f81491e (2026-11-05); awaiting merge and deployment; see INC-01.

Corrections to earlier reports

None.